MIL-C-39014/6B
21 August 1970
SUPERSEDING
MIL-C-39014/6A
1 February 1968

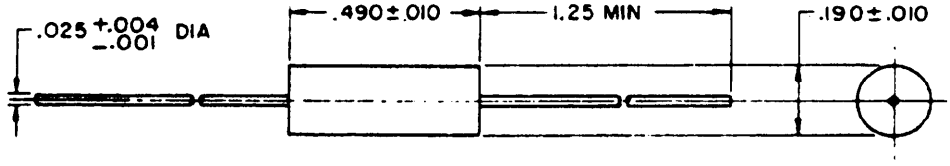
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), ESTABLISHED RELIABILITY STYLE CKR13

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39014.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

INACTIVE FOR NEW DESIGN
FOR REPLACEMENT PURPOSES ONLY



INCHES	MM
.001	.03
.004	.10
.010	.25
.025	.64
.190	4.83
.490	12.45
1.25	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Capacitor, style CKR13

FSC 5910

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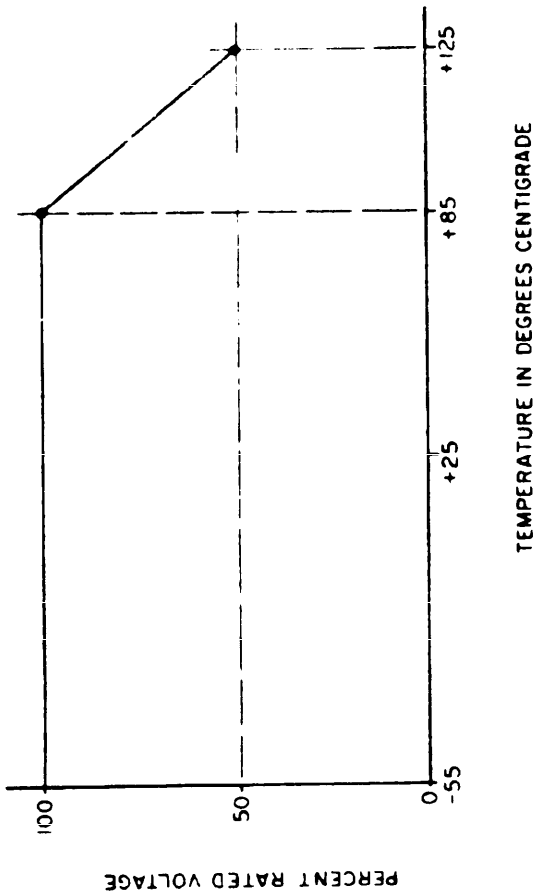


FIGURE 2. Voltage derating with temperature.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Case type: Tubular.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +85°C (125°C with voltage derating, see figure 2).

Voltage rating: 100 volts dc. For operation at 125°C the dc rated voltage shall be derated by 50 percent.

Dissipation factor: Not more than 2.5 percent.

Dielectric withstanding voltage:
Dielectric-Test potential 250 volts dc.

Barometric pressure: Test potential 100 volts dc.

Insulation resistance:
At 25°C - Not less than 100,000 megohms or 1,000 megohm-microfarads, whichever is less.
At 85°C or 125°C - Not less than 10,000 megohms or 100 megohm-microfarads, whichever is less.

Insertion loss: Not applicable.

Seal: Not applicable.

Voltage-temperature limits: In accordance with MIL-C-39014.

Capacitance change with reference to 25°C	
Steps A through D 1/ of table V Bias = 0 volts	Steps E 2/ through G of table V Bias = rated voltage
±15 percent	+15, -25 percent

1/ Step D - At 85°C and 125°C.
2/ Step E - At 85°C with 100 volts dc.
At 125°C with 50 volts dc.

Immersion:
IR - Not less than 50 percent of the initial 25°C requirement.

Salt spray (corrosion): Not applicable.

Moisture resistance: Loading 100 volts dc.

Life:
Rated conditions, 100 percent of dc rated voltage at 125°C.
0 through 2,000 hours:
IR - At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.
ΔC - Less than ±20 percent from initial measured value.
Dissipation factor - Not greater than 2.5 percent.

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4,000 through 32,000 hours:
IR - At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.
ΔC - Less than ±20 percent from initial measured value.
Dissipation factor - Not greater than 2.5 percent.
Accelerated conditions, 200 percent of dc rated voltage at 85°C and 125°C, as applicable.
0 and 250 hours:
IR - At 25°C, not less than 50 percent of initial 25°C requirement.
At 85°C or 125°C, not less than 50 percent of initial 85°C or 125°C requirement.
ΔC - Less than ±20 percent from initial measured value.
Dissipation factor - Not greater than 2.5 percent.
1,000, 2,000, and 4,000 hours:
IR - At 25°C, not less than 15 percent of initial 25°C requirement.
At 85°C or 125°C, not less than 15 percent of initial 85°C or 125°C requirement.
ΔC - Less than ±20 percent from initial measured value.
Dissipation factor - Not greater than 2.5 percent.

Part number: M39014/06-(Dash number from table I).

TABLE I. Dash number and electrical characteristics.

Dash number				Capacitance	Capacitance tolerance
Failure rate level (%/1,000 hours)					
1.0(M)	0.1(P)	0.01(R)	0.001(S)	pF	± percent
0101	0121	0141	0161	15,000	10
0102	0122	0142	0162	15,000	20
0103	0123	0143	0163	22,000	10
0104	0124	0144	0164	22,000	20
0105	0125	0145	0165	33,000	10
0106	0126	0146	0166	33,000	20
0107	0127	0147	0167	47,000	10
0108	0128	0148	0168	47,000	20
0109	0129	0149	0169	68,000	10
0110	0130	0150	0170	68,000	20
0111	0131	0151	0171	82,000	10
0112	0132	0152	0172	82,000	20
0113	0133	0153	0173	100,000	10
0114	0134	0154	0174	100,000	20

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - MI, MU
Navy - EC
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

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